

PROGRAMMING METHODS - LABORATORY 10

Program 01

Use Huffman's algorithm to construct an optimal binary prefix code for the letters shown in the tables below.

Data set a

Letter	Frequency
A	12
B	7
I	18
M	10
S	9
X	5
Z	2

Data set b

Letter	Probability
c	0.11
e	0.22
i	0.16
r	0.12
s	0.15
t	0.10
x	0.14

Description of Huffman's algorithm

```
Huffman()  
  for each symbol, create a tree containing only a root  
  and order the trees according to the probabilities of occurrence of symbols;  
  
  while more than one tree remains:  
    take two trees t1 and t2 with the smallest probabilities p1 and p2 ( $p1 \leq p2$ )  
    create a tree from t1 and t2  
    store  $p1 + p2$  in its root;  
  
  assign 0 to every left branch and 1 to every right branch;  
  
  for each symbol, create a unique code word by moving through the tree  
  from the root to the node containing the given symbol/probability  
  and concatenating the encountered zeros and ones.
```

Laboratory examples

Example 1

Assume the set of characters is {a, b, c, d, e, f}, and every character appears in the file the number of times shown below. Use Huffman's algorithm to construct an optimal binary prefix code for these letters.

Letter	Frequency
a	16
b	5
c	12
d	17
e	10
f	25

Example 2

Assume the set of characters is {A, B, C, D, E}, and their occurrence probabilities are shown below. Use Huffman's algorithm to construct an optimal binary prefix code for these letters.

Letter	Probability
A	0.39
B	0.21
C	0.19
D	0.12
E	0.09

Notes

- Do not use ready-made solutions from the indicated website.
- Apply exception handling.
- Implement your own exception class; for example, use `runtime_error` as a base or reference.

Submission rules

The .cpp file(s) should be named:

Surname_Name_Program_01.cpp

The source files, all other created files, and test text files should be placed in the folder Surname_Name_Laboratory_10.

The folder should be packed as a .rar or .zip archive and submitted to the course folder for Programming Methods - laboratory 10 on delta.pk.edu.pl.

Literature

Neapolitan R.: Foundations of Algorithms with Examples in C++, Helion 2004. Drozd A.: Algorithms and Data Structures, Helion 2004.